

Proof of Concept Pitch: *Program Insights Using Safe, Realistic Synthetic Data*

The purpose of this proof of concept

Your program already collects the information needed to understand resident engagement, stability, and outcomes. What's missing is a way to bring that information together into a clear, actionable picture that supports earlier intervention, stronger programming, and better outcomes.

This proof of concept uses **realistic synthetic data**—not real resident information—to demonstrate exactly what the program could see once the system is connected to live data. It's a safe, privacy-preserving way to show the value before any operational data is touched.

What the synthetic data represents

The synthetic dataset mirrors your actual operational structure:

- **Service encounters** with attendance status
- **Incidents** with severity and type
- **Drug tests** with results and substances
- **Rent charges, payments, waivers, and assistance**
- **Employment snapshots**
- **Stay history and outcomes**
- **Housing structure** (house → room → bed)
- **Referral sources and case managers**
- **Fundraising donors and events**

Every table, field, and relationship in the synthetic dataset corresponds directly to your real OLTP and DW schemas. This means the insights shown in the POC are **structurally accurate**, even though the data itself is fictional.

What the POC will demonstrate

The first version focuses on the program questions that matter most, using synthetic data to illustrate the insights you'll eventually see with real data.

Resident engagement

- Attendance in required vs optional services
- Participation trends over time
- Differences across houses and case managers

Behavioral stability

- Incident patterns and severity
- Drug test positivity trends
- Early-warning indicators based on combined behaviors

Financial responsibility

- Rent compliance
- Waivers and assistance patterns
- Financial stress as a predictor of instability

Employment progress

- Employment status over time
- Wage and hours bands
- Employment at exit

Program outcomes

- Successful vs unsuccessful exits
- Destination types
- Length of stay
- Referral source effectiveness

These insights are generated entirely from synthetic data, but they reflect the **real questions your program can answer** once connected to live data.

What the POC will look like

The dashboard will include five core pages, each populated with synthetic data:

1. **Engagement Overview**

Attendance rates, required vs optional programming, engagement by house and case manager.

2. **Behavior & Stability**

Incident trends, drug test results, combined risk indicators.

3. **Financial Responsibility**

Rent charges vs payments, waivers, assistance, compliance patterns.

4. **Employment Progress**

Employment snapshots, wage bands, hours bands, employment at exit.

5. **Program Outcomes**

Successful exits, destination types, length of stay, referral source performance.

Because the data is synthetic, the POC can be shared freely without privacy concerns.

Why synthetic data strengthens the POC

- **Zero risk** to resident privacy
- **No operational disruption**
- **No need for staff to change workflows**
- **Safe to demo publicly or with funders**
- **Accurate representation** of what real analytics will look like
- **Immediate value** without waiting for data integration

Synthetic data lets leadership see the future system in action without exposing any sensitive information.

What Phase 2 will unlock

Once the director sees the value of the POC, the second version can introduce:

- **Milestones / Achievements** (driver's license, employment obtained, classes completed)
- **Behavioral clustering** (Reliable / Struggling / Chronic)
- **Risk scoring** based on engagement, incidents, drug tests, and rent

- **Case manager performance insights**
- **Predictive early-warning indicators**
- **Longitudinal resident progress models**

Phase 1 shows the vision safely.

Phase 2 delivers the full impact.

“This proof of concept uses realistic synthetic data to show how our existing information can be transformed into a clear picture of resident progress, program effectiveness, and mission impact. It’s a safe, low-risk demonstration of what we can achieve. Once we connect real data, these insights will help us intervene earlier, support residents more effectively, and demonstrate our outcomes with confidence.”